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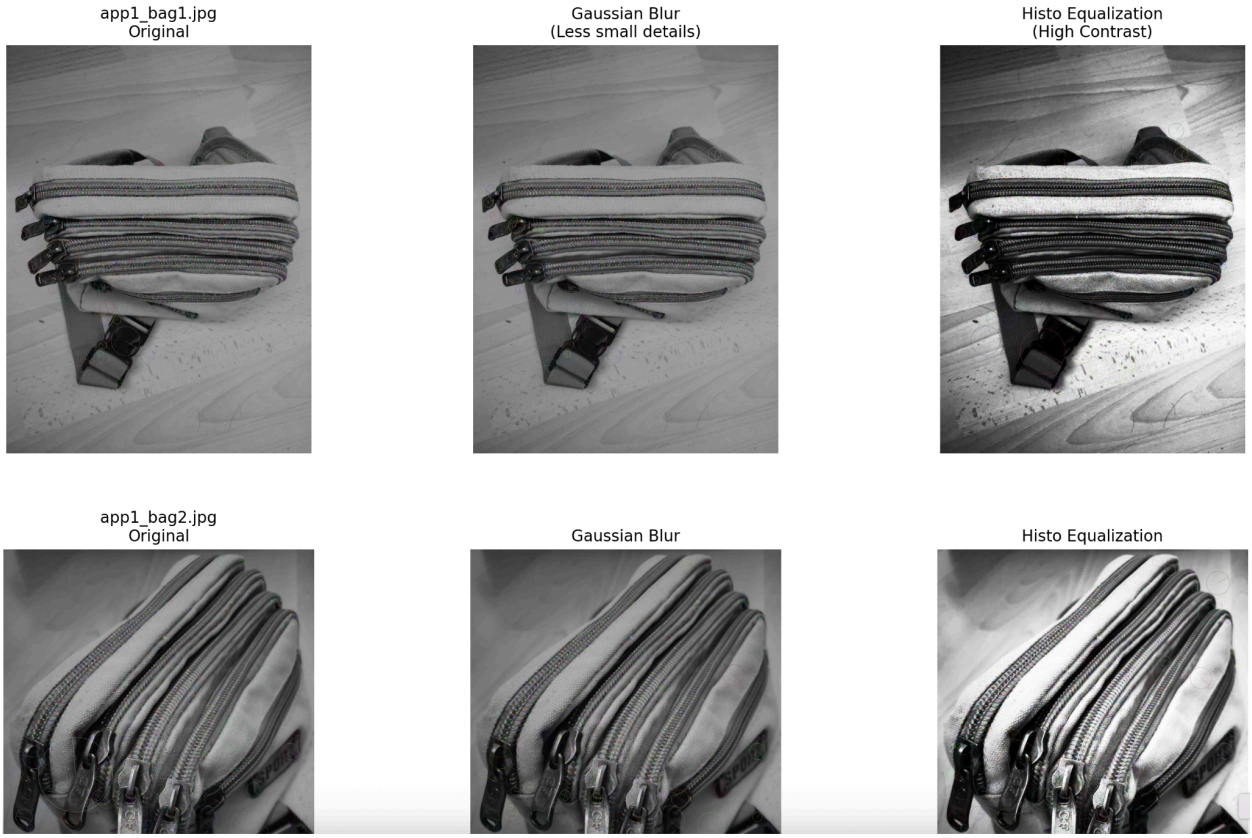
Case 1: SIFT extraction & visualization with opencv

Part 1: Application1 + Questions



Question 1: Where are keypoints concentrated (position, scale, orientation)?

The key points are not uniformly distributed. They appear in regions with strong contrast or complex shapes.

Position: Keypoints are mostly on zipper teeth, bag buckles, letters, and keyboard key corners. These areas have significant changes in brightness.

Scale:

- **Small circles (fine scale):** Appear on high-frequency, repetitive textures such as zipper teeth.
- **Large circles (coarse scale):** Appear on bigger structures like bag corners or whole buckles.

Orientation: Each keypoint includes a dominant orientation represented by a radial line. On the keyboard, orientations are often aligned with the **vertical and horizontal edges** of the keycaps.

Question 2: What happens when the object is texture-poor?

When the object is a flat table surface or smooth surface, the number of detected keypoints drops significantly. These regions lack sharp intensity changes, the Difference of Gaussians (DoG) response is close to zero and the contrastThreshold = 0.04 rejects these low-contrast extrema as unstable.
As a result, such regions remain mostly empty of keypoints.

Question 3: What happens after applying Gaussian blur?

```
[app1_bag1.jpg - Original] Total: 38769 | Strong retained: 11631
[app1_bag1.jpg - Blur] Total: 17704 | Strong retained: 5311
[app1_bag1.jpg - Equalized] Total: 70429 | Strong retained: 21129
[app1_bag2.jpg - Original] Total: 11735 | Strong retained: 3521
[app1_bag2.jpg - Blur] Total: 7117 | Strong retained: 2135
[app1_bag2.jpg - Equalized] Total: 27307 | Strong retained: 8192
[app1_keyboard.jpg - Original] Total: 25184 | Strong retained: 7555
[app1_keyboard.jpg - Blur] Total: 15640 | Strong retained: 4692
[app1_keyboard.jpg - Equalized] Total: 42819 | Strong retained: 12846
```

A Gaussian blur (5×5, σ = 1.0) acts like a **low-pass filter**, affecting SIFT detection.
It significantly reduces the total number of keypoints. Dense clusters on sharp edges (keyboard letters, zipper teeth) disappear or merge.

- SIFT detects extrema in the Difference of Gaussians.
- Blurring removes high-frequency information.
- Smoothed intensity transitions produce weaker gradients and lower DoG responses.

Scale shift:

- Remaining keypoints are generally **larger in scale**.
- Fine textures are lost, but coarse structures (e.g., keycap shapes) remain detectable.

Question 4: What happens after histogram equalization?

```
[app1_bag1.jpg - Original] Total: 38769 | Strong retained: 11631
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```

Histogram equalization stretches the intensity range to [0,255]
It increases the keypoints within previously dark or flat regions. On the bag, uniform fabric shows scattered keypoints. On the keyboard, dark plastic areas between keys gain new features:
Histogram Equalization increases local contrast and previously weak gradients now exceed the contrast threshold. As a result, these regions generate stronger DoG responses. However, sensor noise is also amplified and many newly detected keypoints may be affected by these new noises. This reduces matching reliability across different images.

Question 5: Where is the radius (scale) of a keypoint defined in the code?

- The initial scale is set by **sigma = 1.6.**
- The keypoint radius is not hardcoded.
- During **sift.detectAndCompute()**, each keypoint is assigned a scale.
- The scale is stored in the keypoint's **.size** attribute.
- **drawKeypoints()** uses this attribute to draw circles.

Question 6: Why do keypoint circles differ in size?

SIFT uses a **scale-space pyramid**: Features are detected across progressively blurred and downsampled images.

- **Small circles:** Detected in high-resolution images (fine details).
- **Large circles:** Detected in heavily blurred images (coarse structures).

This allows scale-invariant feature detection.

Part 2: Density & spatial distribution analysis

Question 1: Are keypoints evenly distributed?

No, keypoints are not evenly distributed because SIFT is content-aware. Dense "groups" of keypoints appear on textured regions (zipper, keyboard letters). Large uniform regions (table, smooth fabric) remain empty.

Question 2: Are keypoints concentrated on texture?

Yes, SIFT relies on DoG, which responds to high-frequency gradients and texture corresponds to rapid intensity changes.

- In the keyboard image: contrast between white letters and black keycaps.
- In the bag image: repetitive zipper teeth generate dense keypoint clusters.

Question 3: What happens in uniform regions?

In uniform regions, pixel values are nearly identical, DoG response is close to zero and contrastThreshold = 0.04 rejects these regions.

This prevents detecting unstable points caused by noise.

Question 4: Observations on images with repetitive patterns (e.g., buildings with windows)

Repetitive structures create challenges such as:

- **High density:** Each repeated element triggers similar keypoints.
- **Ambiguity:** Descriptors of neighboring elements are almost identical.
- **Geometric regularity:** Grid-like keypoint patterns appear for example keyboard layout

While detection is easy, matching becomes difficult due to lack of uniqueness.